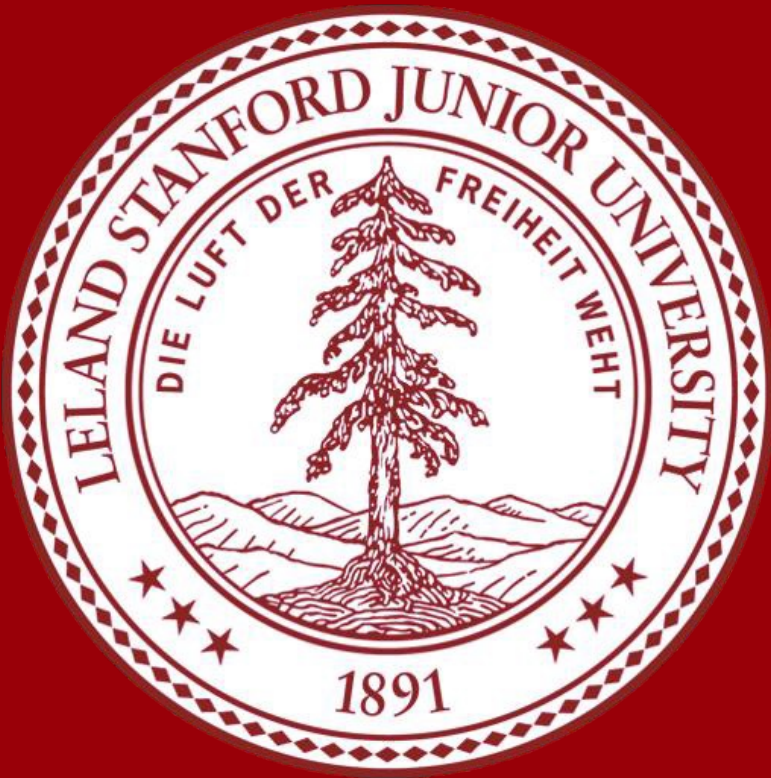




Addressing Gradient Filtering in Differentiable Repair Layers for Neural Optimization

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Differentiable Repair Layers

Parametric Constrained Optimization

$$\mathcal{P}(x) : \underset{y}{\text{minimize}} \varphi(y, x) \text{ subject to } y \in \mathcal{C}(x)$$

Hard-constrained Neural Networks

Goal : Amortize the heavy computational cost of traditional solvers into a rapid forward pass for repetitive parametric solves.

Architecture: A neural backbone f_θ maps parameters x to a raw prediction $\hat{y}$.

Feasibility-by-Design: A parameter-free repair layer $\mathcal{R}$ projects (or iteratively repairs) the infeasible $\hat{y}$ onto the constraint set $\mathcal{C}(x)$ to guarantee a feasible output $\tilde{y}$.

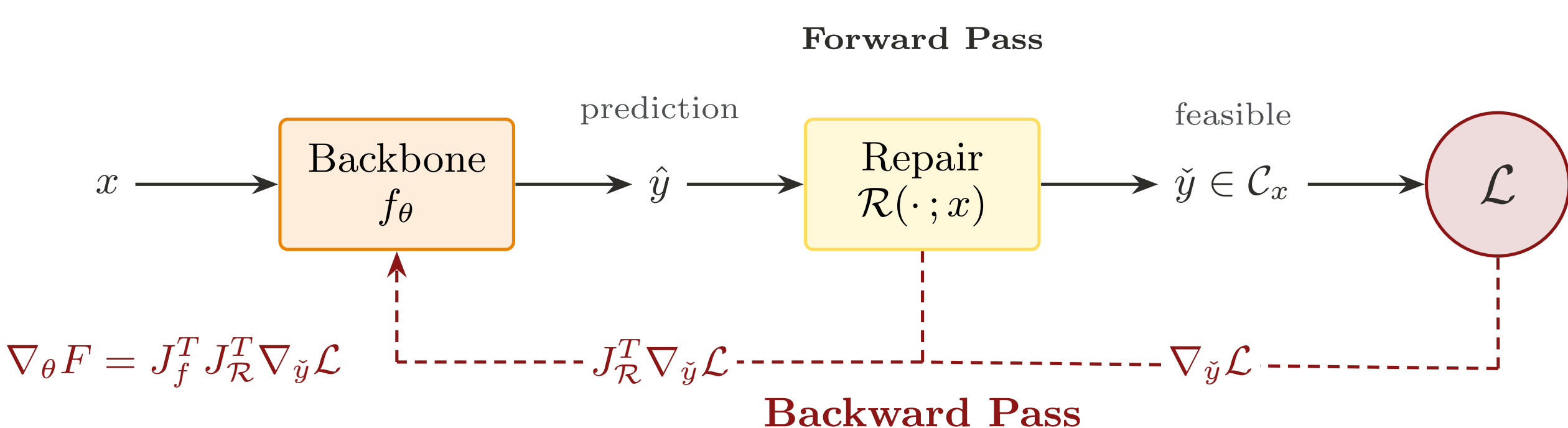


Figure 1: Schematic view of the general architecture of a Hard-constrained Neural Network.

Gradient Filtering

Special Case of Projection-based Repair

For projection-based repair layers, the Jacobian $J_{\mathcal{R}}$ acts as a restrictive **geometric filter** on the backpropagated task gradient:

- **Normal-Space Annihilation** : Gradient components orthogonal to the active constraints are zeroed.
- **Distance-Dependent Attenuation**: Gradient components tangent to the manifold are scaled down by a factor $s_k = \frac{1}{1+d \cdot \lambda_k}$

Repair Delegation

Consequences of Gradient Filtering

- Performance instability of **repair layer hyperparameters** (step size, iterations ...)
- **Training collapse** as gradient vanish (convex constraint) or explode (concave constraint)
- Increased **repair overhead** (forward & backward pass)

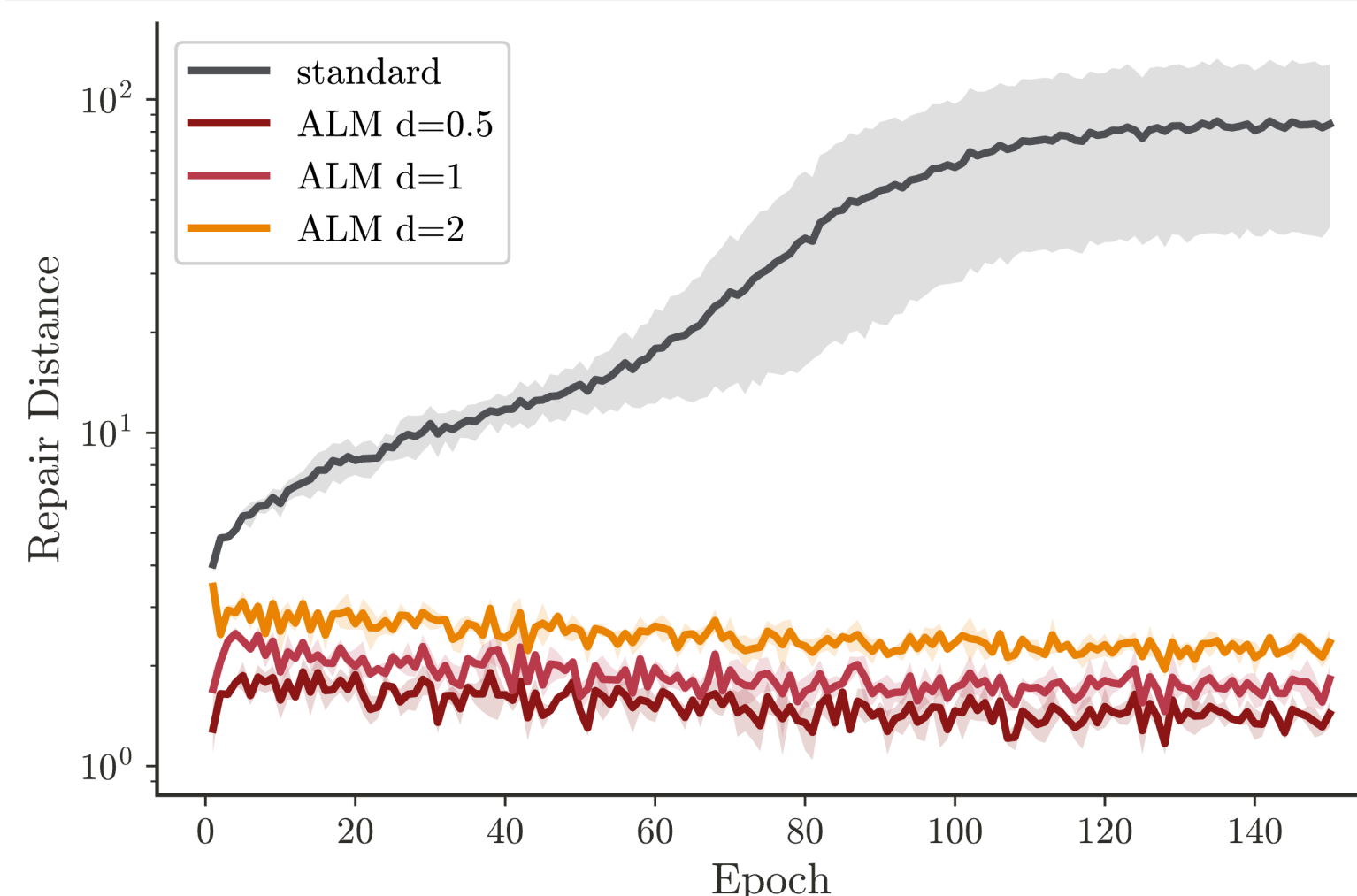


Figure 2: Repair distance grows exponentially until collapse in standard training.

Solving with ALM

Formulation

$$\begin{aligned} \min_{\theta} \mathbb{E}_x [\mathcal{L}(\mathcal{R}(f_\theta(x); x))] \\ \text{s.t. } \mathbb{E}_x [\|f_\theta(x) - \mathcal{R}(f_\theta(x); x)\|_2^2] \leq d \end{aligned}$$

- **Augmented Lagrangian Method (ALM)** :

1. Gradient step on θ using the augmented loss for batch i :

$$\mathcal{L} = \underbrace{\mathbb{E}[\mathcal{L}(\mathcal{R}(f_\theta))]_{\text{task loss}}} + \underbrace{\mathbb{E}\left[\frac{1}{2\rho} \max(0, \lambda + \rho(d_i - d))^2\right]}_{\text{ALM penalty}}$$

2. Update the multiplier from the observed violation:

$$\lambda \leftarrow \max(0, \lambda + \alpha_\lambda(d_i - d))$$

- The additional gradient term introduced is **orthogonal** to the current one (though not necessarily in the parameter-space) and is computed with **zero overhead**.

Empirical Results

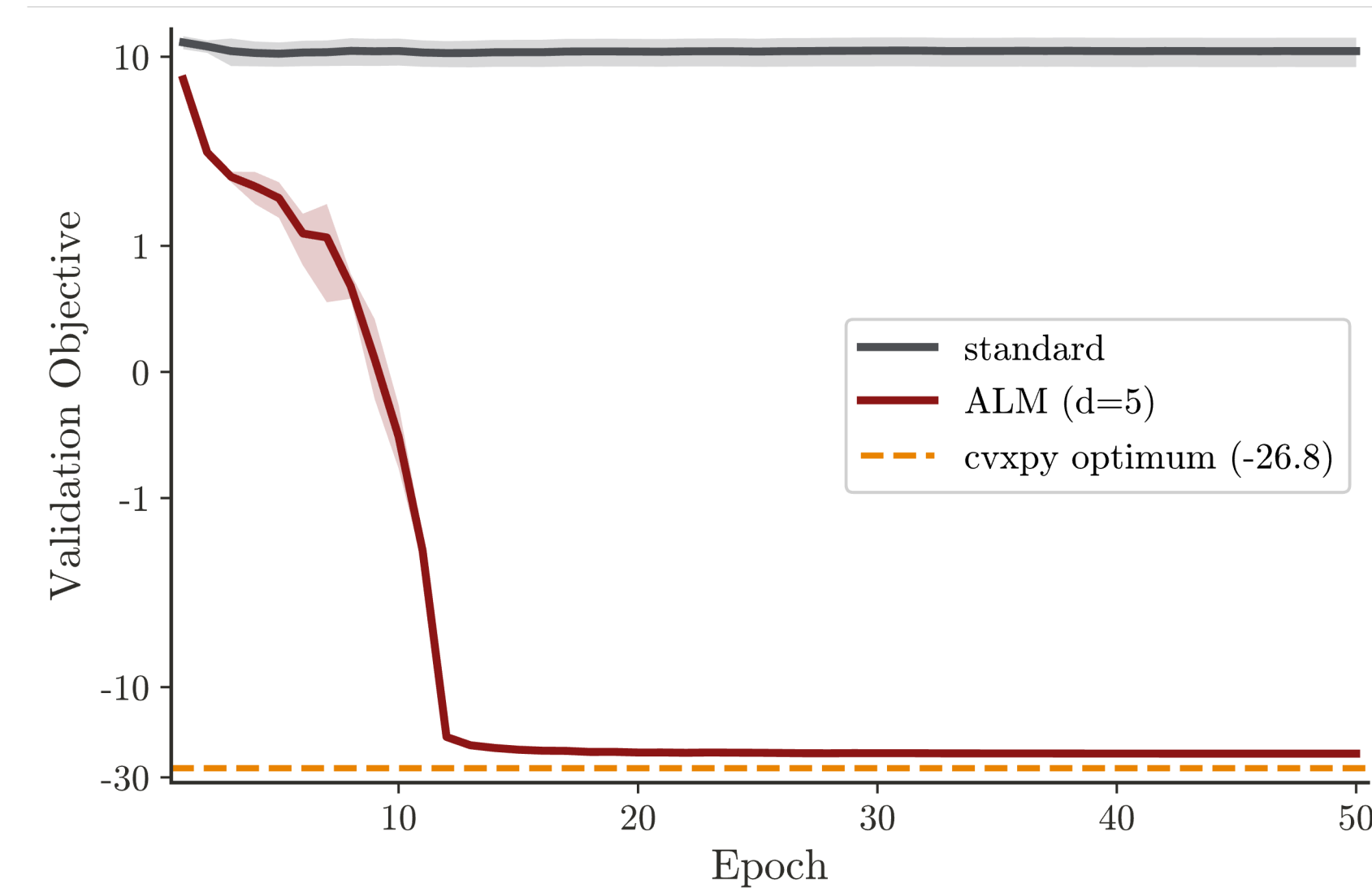


Figure 3: ALM enables to reach optimal solutions in challenging settings where the standard method fails

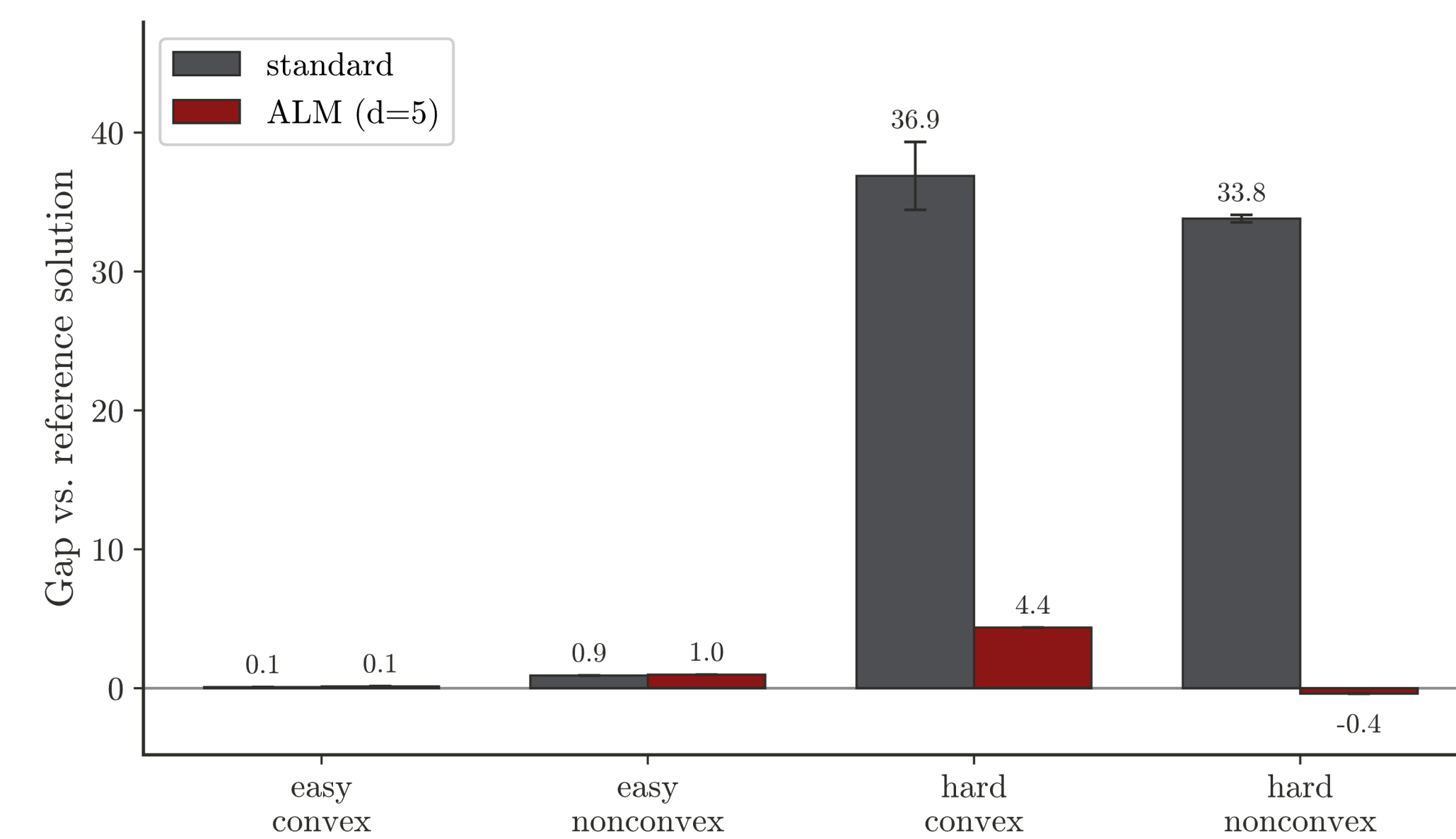


Figure 4: ALM preserves performance on easy instances and provides significant improvements on harder instances

Additional practical benefits observed in practice :

- **Better feasibility guarantees** for iterative repairs as the number of iterations to converge stays **bounded**.
- ALM reaches **diverse solutions** and avoids **implicit bias** (e.g. towards large active sets) of long-distance repairs.

References

- Chu et al. (2026) *SnareNet*. arXiv:2602.09317.
- Bonnans & Shapiro. (2000) *Perturbation Analysis of Opt. Problems*. Springer.
- Nguyen & Donti. (2025) *FSNet*. arXiv:2506.00362.